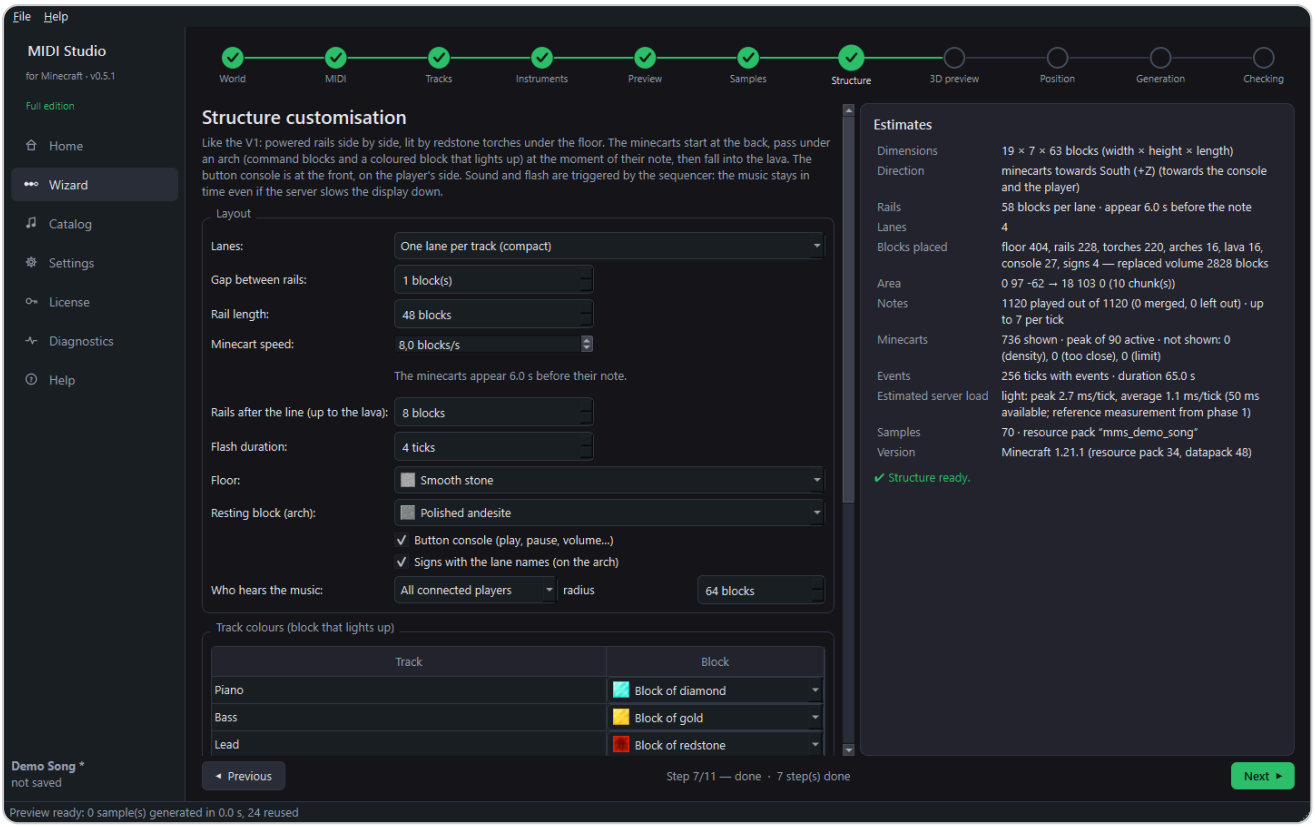
Step 8 — The structure (wizard step 7)

Goal: choose how the music looks in the world.



Structure settings

What gets built

- **Powered rails side by side**, lit by redstone torches under the floor.
- The minecarts appear at the far end and travel towards you.
- Above the **trigger line**: an arch of command blocks and a **coloured block that lights up** when the note plays (one colour per track).
- After the line, the minecarts **fall into a lava pit**.
- At the front, on your side: a **button console** (play, pause, stop, volume −/+, tempo −/+, pitch −/+) with signs.

The sound and the flashes are triggered by the datapack's sequencer: the minecarts are only visual, so the music stays in time even if the server slows down.

Layout settings

- **Lanes** — one per track (compact), one per played note (piano roll) or one per pitch.
- **Gap between rails** — 1 = rails right next to each other.
- **Rail length** and **Minecart speed** — they decide how long before its note a minecart appears.
- **Rails after the line** — distance to the lava.
- **Floor**, **Resting block** of the arch, **Flash duration**, **Track colours**.
- **Button console**, **Signs with the lane names**, **Who hears the music** (everyone, or players near the structure).

Optimisation

Choose a profile: **Best quality**, **Balanced** or **Performance**. *Custom* (full edition) unlocks every setting: maximum active minecarts, minecart density, minecart reuse, simultaneous notes per tick… The **Estimates** panel shows the size, the number of blocks, the minecarts and the estimated server load in real time.

Next guide: 3D preview and position.